

RRC_Raptor_V4H

- IPL_Porting Guide

V0.0.1

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i.VERSION LIST

Description	Version	Date	Author
Porting Guide	V0.0.1	2024/08/05	Renny Hsu
Modify compile step.		2024/09/18	Renny Hsu
Modify DIP switch setting.		2024/09/18	Renny Hsu

ii.HASHTAG

#V4H
#IPL
#ICUMX
#CX 2nd
#Flash Writer

1. Overview

The IPL (Initial Program Loader) include three parts : ICUMX IPL, CX 2nd IPL, and Flash Writer.

1.1 ICUMX (Intelligent Cryptographic Unit/Master Extension) IPL

ICUMX IPL is loaded and executed by the BootROM program. ICUMX IPL loads images from external flash memory and executes an integrity check loaded images using a certificate.

1.2 Dummy FW (Secure FW)

After ICUMX IPL executed, the processor ICUMX performed firmware security services.

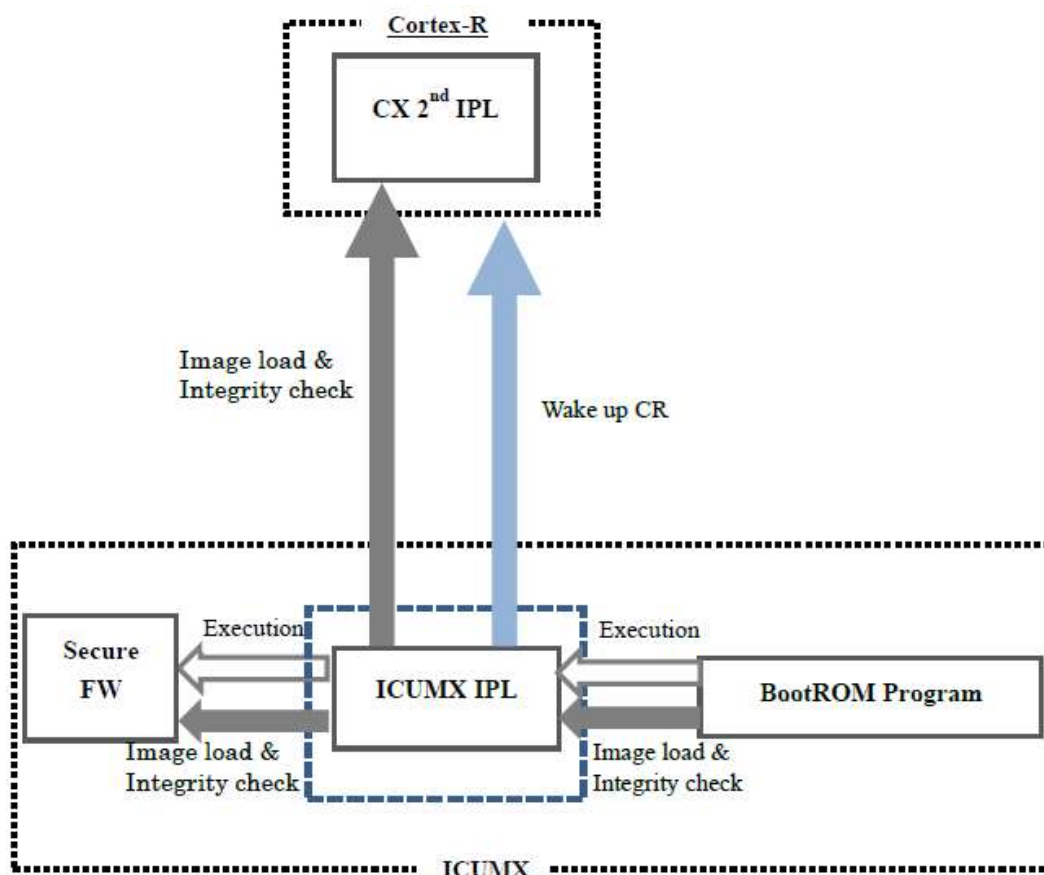


Figure 1-1. Scope of the Internal Program of ICUMX IPL.

1.3 CX 2nd IPL

CX 2nd IPL is loaded and executed by the ICUMX IPL program. CX 2nd IPL loads binary images (Secure Monitor, Trusted OS and Non-Secure OS) from external flash memory. After

CX 2nd IPL loads the binary images, it executes the binary image of Secure Monitor.

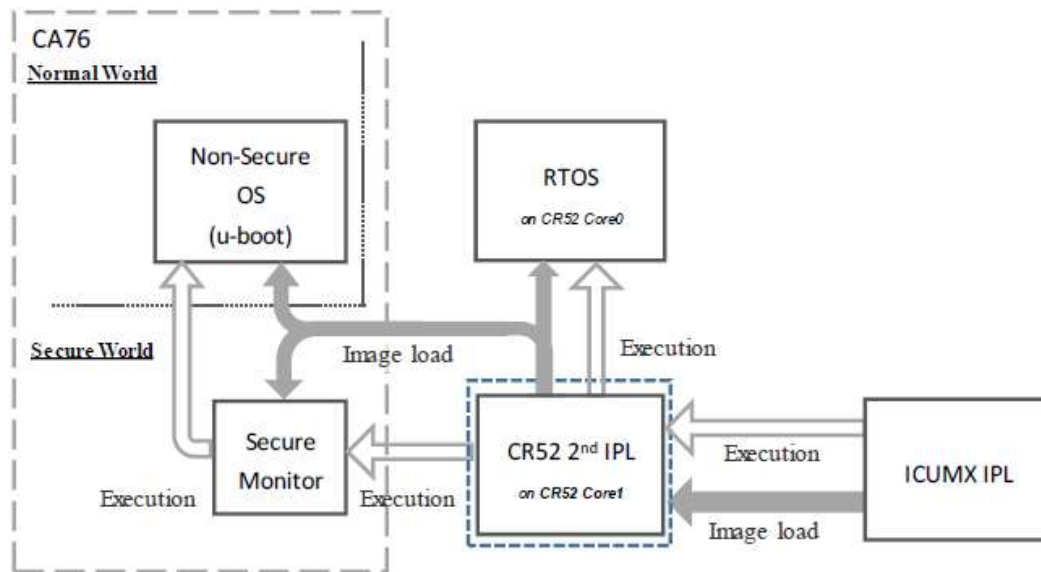


Figure 1-2. Scope of the Internal Program of CX 2nd IPL.

1.4 Flash Writer

The Flash writer is downloaded from the Host PC via SCIF by Boot ROM.

1.5 Secure Monitor and U-boot

Those programs are running on Cortex CA76.

2. Compile Environment

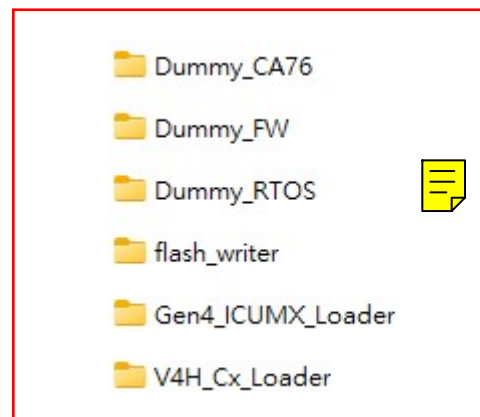


Figure 2-1. Source code list.

2.1. [ICUMX IPL / SA0 / SA9] :

- Operation system : Windows 10 or Windows 11
- Compiler : Green Hills Software V800 Development Tool (v2023.1)
- Terminal : Cygwin (Cygwin Version : 2.10.0-1, Make Version : 4.2.1.2)
- Source code folder : Gen4_ICUMX_Loader

2.2. [Secure FW] :

- Operation system : Windows 10 or Windows 11
- Compiler : Green Hills Software V800 Development Tool (v2023.1)
- Terminal : Cygwin (Cygwin Version : 2.10.0-1, Make Version : 4.2.1.2)
- Source code folder : Dummy_FW

2.3. [CX 2nd IPL] :

- Operation system: Ubuntu 20.04 LTS (64bit)
- Compiler : gcc-arm-10.3-2021.07-x86_64-arm-none-eabi.tar.xz
- Source code folder : V4H_Cx_Loader

2.4. [Dummy RTOS] :

- Operation system: Ubuntu 20.04 LTS (64bit)
- Compiler : gcc-arm-10.3-2021.07-x86_64-arm-none-eabi.tar.xz
- Source code folder : Dummy_RTOS

2.5. [Flash Writer] :

- Operation system : Windows 10 or Windows 11
- Compiler : Green Hills Software V800 Development Tool (v2023.1)
- Terminal : Cygwin (Cygwin Version : 2.10.0-1, Make Version : 4.2.1.2)
- Source code folder : flash_writer

2.6. [Secure Monitor(BL31) / U-boot] :

- Operation system: Ubuntu 20.04 LTS (64bit)
- Compiler : gcc-arm-10.3-2021.07-x86_64-aarch64-none-elf.tar.xz
- Source code folder : Dummy_CA76

3. Install compile and download tool

3.1. Cygwin : (Windows OS)

Download the Cygwin installer from the site shown below.

Cygwin (<https://www.cygwin.com/>)

Launch the downloaded installer and install according to the instructions of the installer. Install at least the package "make: The GNU version of the 'make' utility" always.

3.2. Green Hills Software V800 Development Tool (v2023.1) : (Windows OS)

For ICUMX IPL / SA0 / SA9, Flash Writer and Secure FW, please mount GHS_MULTI_814_202314.iso and reference installation.pdf, Chapter 2.

3.3. Cortex R52 GCC compiler : (Ubuntu 20.04)

For CX 2nd IPL and Dummy RTOS, prepare the GCC cross compiler. Please refer to r11uz0236ej0021-icumx-ipl.pdf, Chapter 6.2.1.

```
$ wget https://developer.arm.com/-/media/Files/downloads/gnu-rm/10.3-2021.10/gcc-arm-none-eabi-10.3-2021.10-x86_64-linux.tar.bz2
$ tar jxvf gcc-arm-none-eabi-10.3-2021.10-x86_64-linux.tar.bz2
$ export PATH=$PATH:{GCC cross compiler unzip folder}/gcc-arm-none-eabi-10.3-2021.10/bin
```

3.4. Cortex A76 GCC compiler : (Ubuntu 20.04)

For Secure Monitor(BL31) and U-boot, prepare the GCC cross compiler. Please refer to r11uz0236ej0021-icumx-ipl.pdf, Chapter 6.2.1.

```
$ wget https://developer.arm.com/-/media/Files/downloads/gnu-a/10.3-2021.07/binrel/gcc-arm-10.3-2021.07-x86_64-aarch64-none-elf.tar.xz
$ tar Jxvf gcc-arm-10.3-2021.07-x86_64-aarch64-none-elf.tar.xz
$ export PATH=$PATH:{GCC cross compiler unzip folder}/gcc-arm-10.3-2021.07-x86_64-aarch64-none-elf/bin
```

3.5. PuTTY : (Ubuntu 20.04)

For download all firmware.

```
$sudo apt install -y putty
```

4. Compile Step

4.1. ICUMX IPL and CX 2nd IPL :

4.1.1. If ICUMX IPL loaded from QSPI flash on 40MHz, Dummy RTOS, Secure

Monitor(BL31) and U-boot loaded from eMMC.

1. Check RCAR_SA9_TYPE = 1

```
$vi Gen4_ICUMX_Loader/Makefile  
Line 373 : RCAR_SA9_TYPE := 1
```

2. Execute Cygwin64 Terminal in Windows OS.

```
$ cd Gen4_ICUMX_Loader/  
$ make clean  
$ make LSI=V4H DEBUG=DISABLE BOARD=V4H_SOM  
APP_LOAD_BY=CX_2ND_LOADER APP_PROG_LOAD=EMMC  
QSPI_FLASH_MODE=SDR
```

When the build is completed, the following binary image is output :

```
./build/release/bootparam_sa0.srec (Used only when Normal Boot)  
./build/release/cert_header_sa9.srec (Used only when Normal Boot)  
./build/release/icumx_loader.srec
```

3. Execute Terminal in Ubuntu OS

```
$ cd V4H_Cx_Loader/  
$ make clean  
$ make LSI=V4H BOOT_MODE=NORMAL BOARD=V4H_SOM  
APP_LOAD_BY=CX_2ND_LOADER APP_PROG_LOAD=EMMC  
QSPI_FLASH_MODE=SDR
```

When the build is completed, the following binary image is output :

```
./build/release/cr52_loader.srec
```

4.1.2. If ICUMX IPL, CX 2nd IPL, Dummy RTOS, Secure Monitor(BL31) and U-boot

loaded from QSPI flash SDR mode on 40MHz.

1. Check RCAR_SA9_TYPE = 1

```
$ vi Gen4_ICUMX_Loader/Makefile  
Line 373 : RCAR_SA9_TYPE := 1
```

2. Execute Cygwin64 Terminal in Windows OS.

```
$ cd Gen4_ICUMX_Loader/  
$ make clean  
$ make LSI=V4H DEBUG=DISABLE BOARD=V4H_SOM  
APP_LOAD_BY=CX_2ND_LOADER APP_PROG_LOAD=QSPI_40M  
QSPI_FLASH_MODE=SDR
```

When the build is completed, the following binary image is output :

```
./build/release/bootparam_sa0.srec (Used only when Normal Boot)  
./build/release/cert_header_sa9.srec (Used only when Normal Boot)  
./build/release/icumx_loader.srec
```

3. Execute Terminal in Ubuntu OS.

```
$ cd V4H_Cx_Loader/  
$ make clean  
$ make LSI=V4H BOOT_MODE=NORMAL BOARD=V4H_SOM  
APP_LOAD_BY=CX_2ND_LOADER APP_PROG_LOAD=QSPI_40M  
QSPI_FLASH_MODE=SDR
```

When the build is completed, the following binary image is output :

```
./build/release/cr52_loader.srec
```

- 4.1.3. If ICUMX IPL, CX 2nd IPL, Dummy RTOS, Secure Monitor(BL31) and U-boot

loaded from eMMC.

1. Check RCAR_SA9_TYPE = 0

```
$ vi Gen4_ICUMX_Loader/Makefile  
Line 373 : RCAR_SA9_TYPE := 0
```

2. Execute Cygwin64 Terminal in Windows OS.

```
$ cd Gen4_ICUMX_Loader/
```



```
$ make clean
$ make LSI=V4H DEBUG=DISABLE BOARD=V4H_SOM
APP_LOAD_BY=CX_2ND_LOADER APP_PROG_LOAD=EMMC
QSPI_FLASH_MODE=SDR
```

When the build is completed, the following binary image is output :

```
./build/release/bootparam_sa0.srec (Used only when Normal Boot)
./build/release/cert_header_sa9.srec (Used only when Normal Boot)
./build/release/icumx_loader.srec
```

3. Execute Terminal in Ubuntu OS

```
$ cd V4H_Cx_Loader/
$ make clean
$ make LSI=V4H BOOT_MODE=NORMAL BOARD=V4H_SOM
APP_LOAD_BY=CX_2ND_LOADER APP_PROG_LOAD=EMMC
QSPI_FLASH_MODE=SDR
```

When the build is completed, the following binary image is output :

```
./build/release/cr52_loader.srec
```

4.1.4. If ICUMX IPL, CX 2nd IPL, Dummy RTOS, Secure Monitor(BL31) and U-boot

loaded from QSPI flash DDR mode on 80MHz.

1. Check RCAR_SA9_TYPE = 1

```
$ vi Gen4_ICUMX_Loader/Makefile
Line 373 : RCAR_SA9_TYPE := 1
```

2. Execute Cygwin64 Terminal in Windows OS.

```
$ cd Gen4_ICUMX_Loader/
$ make clean
$ make LSI=V4H DEBUG=DISABLE BOARD=V4H_SOM
APP_LOAD_BY=CX_2ND_LOADER APP_PROG_LOAD=QSPI_80M
QSPI_FLASH_MODE=DDR
```

When the build is completed, the following binary image is output :

```
./build/release/bootparam_sa0.srec (Used only when Normal Boot)
./build/release/cert_header_sa9.srec (Used only when Normal Boot)
./build/release/icumx_loader.srec
```

3. Execute Terminal in Ubuntu OS.

```
$ cd V4H_Cx_Loader/  
$ make clean  
$ make LSI=V4H BOOT_MODE=NORMAL BOARD=V4H_SOM  
APP_LOAD_BY=CX_2ND_LOADER APP_PROG_LOAD=QSPI_80M  
QSPI_FLASH_MODE=DDR
```

When the build is completed, the following binary image is output :

./build/release/cr52_loader.srec

4.2. Flash Writer :

```
Execute Cygwin64 Terminal in Windows OS,  
$ cd flash_writer/src/v4h/  
$ make clean  
$ make LSI=V4H WORK=SDRAM
```

When the build is completed, the following binary image is output :

./sec/v4h/ICUMX_output/ICUMX_Flash_writer_SCIF_DUMMY_CERT_EB203000_V4H
.mot

4.3. Secure FW (Dummy FW) :

```
Execute Cygwin64 Terminal in Windows OS,  
$ cd Dummy_FW/  
$ make LSI=V4H DEBUG=DISABLE
```

When the build is completed, the following binary image is output :

./build/release/dummy_fw.srec

4.4. Dummy RTOS :

```
Execute Terminal in Ubuntu OS,  
$ cd Dummy_RTOS/  
$ make clean  
$ make
```

When the build is completed, the following binary image is output :

./build/release/dummy_rtos.srec

4.5. Secure Monitor(BL31) and U-boot :

```
Execute Terminal in Ubuntu OS,  
$ cd Dummy_CA76/  
$ make clean  
$ make DUMMY=1  
$ make DUMMY=2
```

*DUMMY=1 is Secure Monitor

*DUMMY=2 is U-boot

When the build is completed, the following binary image is output :

./AArch64_output/AArch64_Dummy_CA76_Program.srec is Secure Monitor

./AArch64_output/AArch64_Dummy_CA76_Program2.srec is U-boot

5. Firmware download

5.1. Power OFF.

5.2. Other DIP switch setting :

Table 5-1. SW26 ~ SW32 default.

	Pin1	Pin2	Pin3	Pin4	Pin5	Pin6	Pin7	Pin8
SW26	ON	ON	ON	ON	ON	ON	ON	ON
SW35	ON	ON	ON	ON	ON	ON	ON	ON
SW31	ON	ON	ON	ON	ON	ON	ON	ON
SW27	ON	ON	ON	ON	ON	ON	ON	ON
SW30	ON	ON	ON	ON	ON	ON	ON	ON
SW32	ON	ON	ON	ON	ON	ON	ON	ON

Table 5-2. SW28 definition and default.

SW28	Pin1	Pin2	Pin3	Pin4	Pin5	Pin6	Pin7	Pin8
Definition	MD7	MD6	-	MD8	MD4	MD3	MD2	MD1
Default	OFF	ON	OFF	OFF	OFF	OFF	OFF	OFF

Table 5-3. SW29 definition and default.

SW29	Pin1	Pin2	Pin3	Pin4	Pin5	Pin6	Pin7	Pin8
Definition	MD27	MD22	-	-	-	-	-	-
Default	ON	ON	ON	ON	OFF	ON	ON	ON

Table 5-4. SW32 definition and default.

SW32	Pin1	Pin2	Pin3	Pin4	Pin5	Pin6	Pin7	Pin8
Definition	MD37	MD36	MD25	MD23	MD33	MD24	MD18	MD9
Default	ON	ON	ON	ON	ON	ON	ON	ON

Table 5-5. SW33 definition and default.

SW33	Pin1	Pin2	Pin3	Pin4	Pin5	Pin6	Pin7	Pin8
Definition	MD14	MD13	MD29	MD32	MD31	-	MD16	-
Default	ON	ON	ON	ON	OFF	ON	OFF	OFF-

Table 5-6. SW36 definition and default.

SW36	Pin1	Pin2	Pin3	Pin4	Pin5	Pin6	Pin7	Pin8
------	------	------	------	------	------	------	------	------

Definition	MD5	MD35	MDT0	MD21	MD20	MD11	MD10	-
Default	OFF	ON	ON	ON	OFF	ON	OFF	OFF

Table 5-7. SW34 definition and default.

SW34	Pin1	Pin2
Definition	MD25	SL_SW2
Default	OFF	OFF

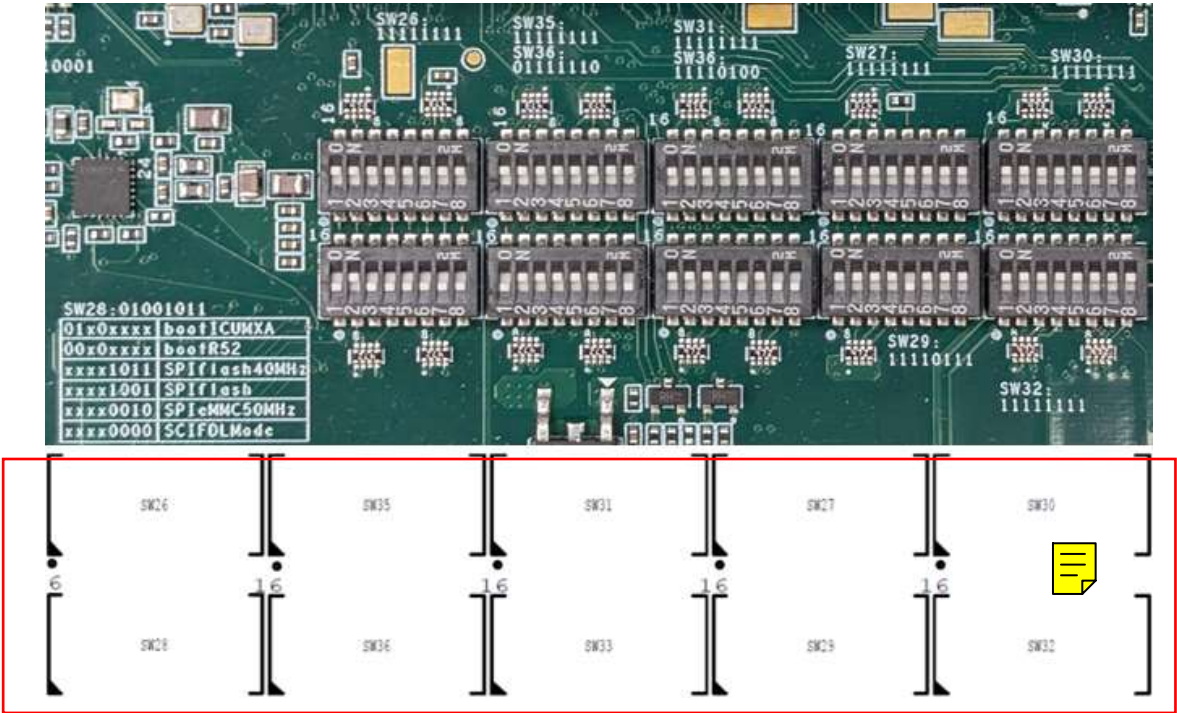


Figure 5-1. Default of DIP switches.

5.3. Prepare download firmware :

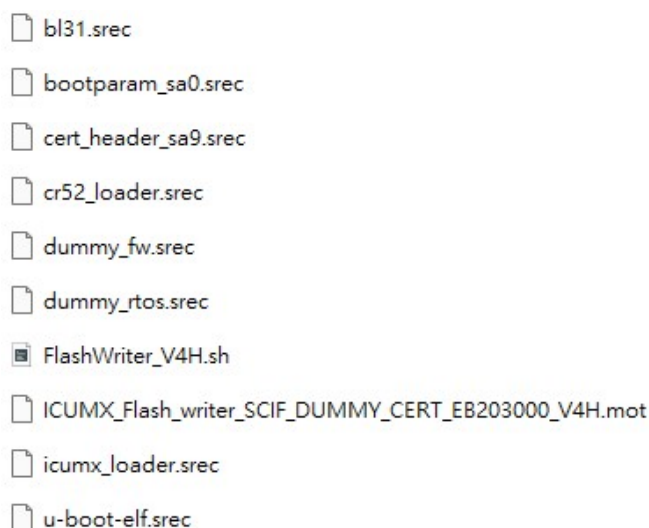


Figure 5-2. Download firmware.

Table 5-8. Information list of release images for V4H when ICUMX IPL loads CX 2nd IPL (QSPI Flash).

File name	Loaded by	Program Top Address	Flash Save Address	Description
bootparam_sa0.srec	BootROM	H'EB20 0000	H'0000 0000	ICUMX IPL (Boot parameter)
icumx_loader.srec	BootROM	H'EB21 0000	H'0004 0000	ICUMX IPL
cert_header_sa9.srec	ICUMX IPL	H'EB23 0000	H'0024 0000	ICUMX IPL (Certificate)
dummy_fw.srec	ICUMX IPL	H'EB24 0000	H'0028 0000	Dummy firmware (Instead of secure firmware)
cr52_loader.srec	ICUMX IPL	H'E630 0000	H'0048 0000	CX 2 nd IPL

Table 5-9. Information list of release images for V4H when ICUMX IPL loads CX 2nd IPL (eMMC).

File name	Loaded by	Program Top Address	eMMC Save Address	eMMC Save Sectors*1	Description
dummy_rtos.srec	CX 2 nd IPL	H'E210 0000	boot partition 1	H'0D00	Dummy RTOS

					(Instead of CR main OS)
bl31.srec	CX 2 nd IPL	H'4640 0000	boot partition 1	H'8D00	Dummy Secure monitor (Instead of Secure monitor)
u-boot-elf.srec	CX 2 nd IPL	H'5000 0000	boot partition 1	H'9500	Dummy U-boot

Note *1) 1 sector is 512 bytes.

5.4. Connect USB Host connector of Ubuntu Host PC that is virtual COM port to

Carried Board(CN23) with USB cable for displaying console.

5.5. Check the USB device node exist:

```
rennyhsu@ubuntu:~$ ls -l /dev/ttyUSB0
crw-rw---- 1 root dialout 188, 0  6月 24 10:50 /dev/ttyUSB0
```

Figure 5-3. Check USB device.

5.6. Activate the Terminal Software PuTTY on Ubuntu Host PC. Configure the

PuTTY on Ubuntu Host PC as followings:

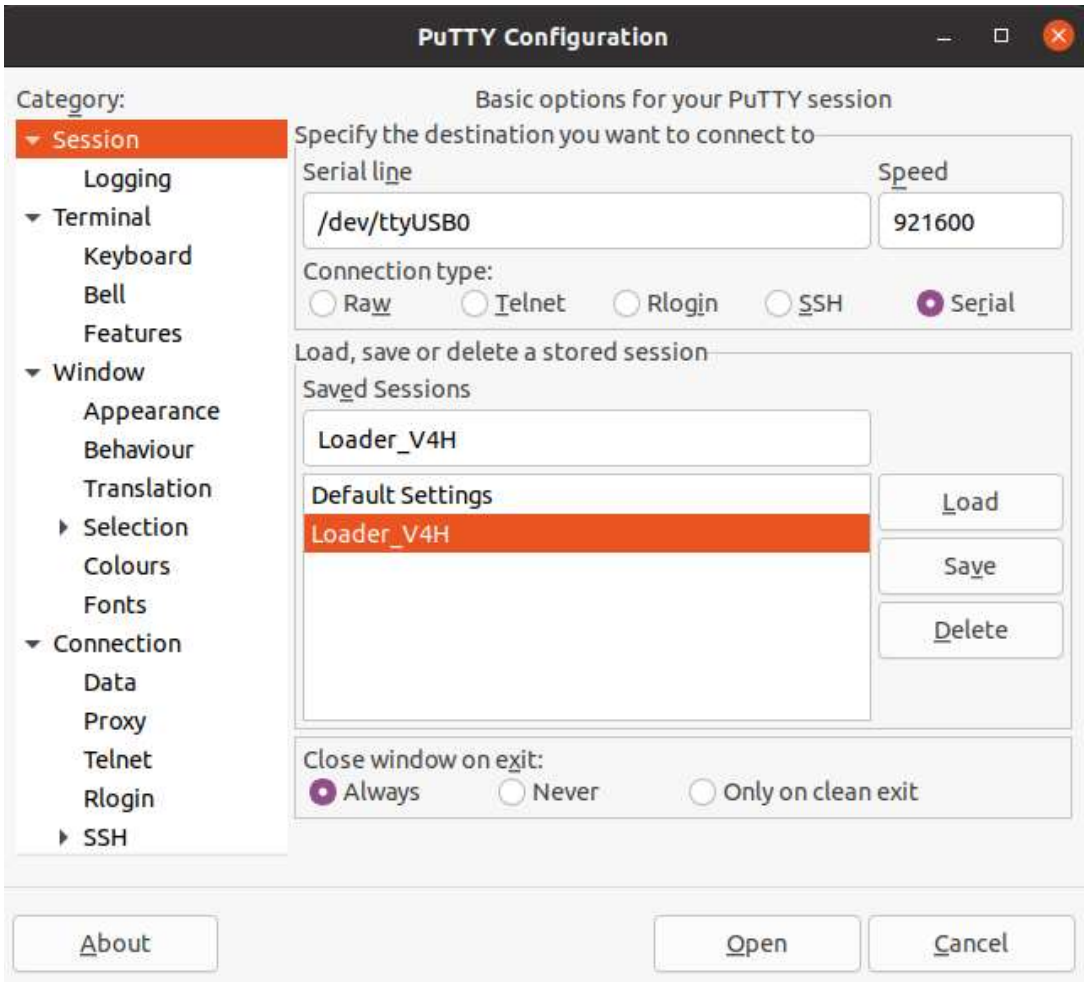


Figure 5-4. PuTTY setting.

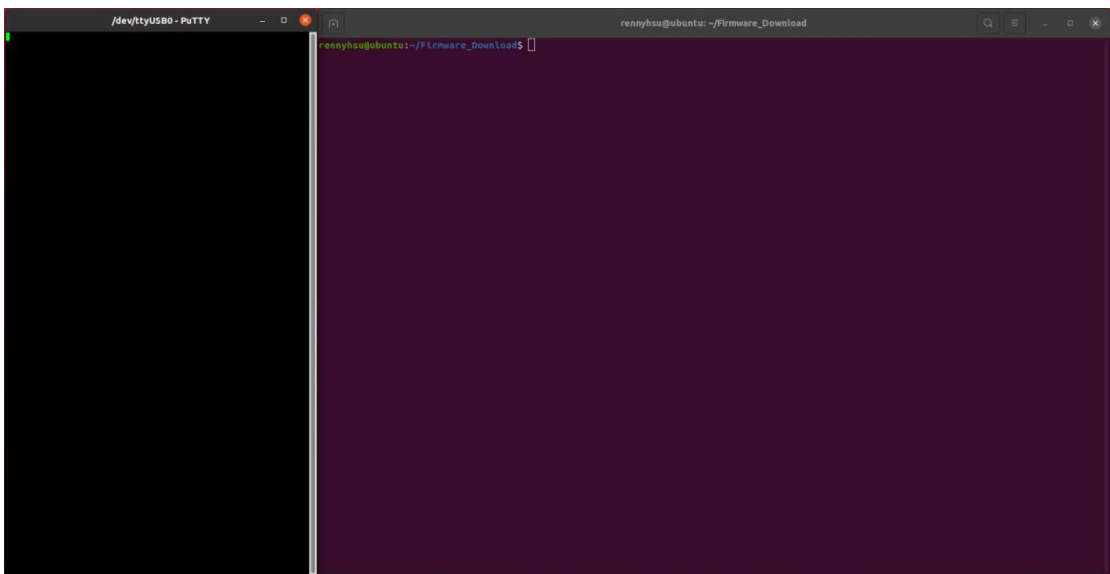


Figure 5-5. PuTTY and terminal.

5.7. Check SW28 setting:

Table 5-10. SCIF/HSCIF download mode of SW28

	Pin1	Pin2	Pin3	Pin4	Pin5	Pin6	Pin7	Pin8
SW28	OFF	ON	OFF	OFF	OFF	OFF	OFF	OFF

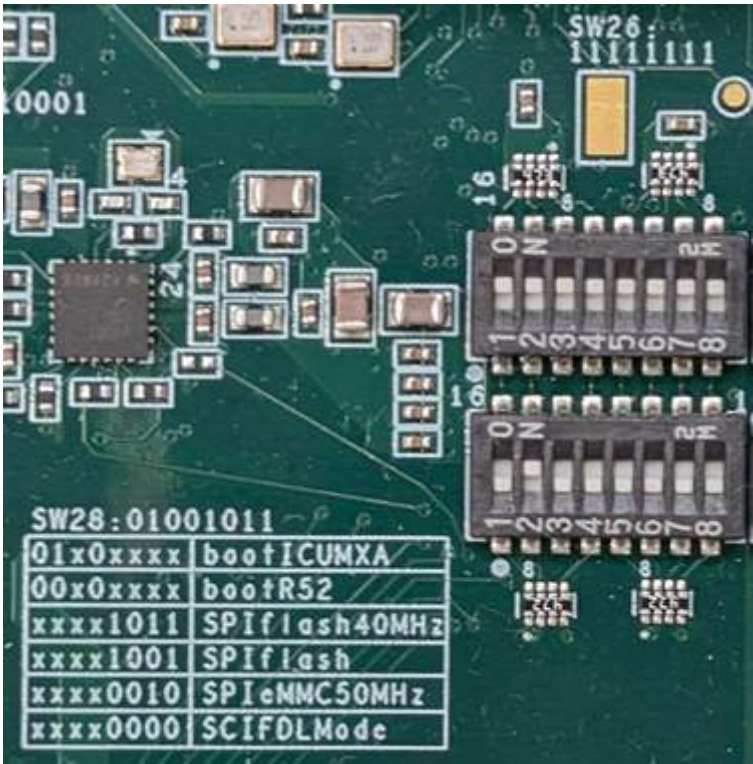


Figure 5-6. SCIF/HSCIF download mode of SW28.

5.8. The SCIF download mode message will be output after the power on.



Figure 5-7. SCIF download mode message.

5.9. Execute “FlashWriter_V4H.sh” download firmware to QSPI Flash and eMMC.

Table 5-11. Flash partition list.

Partition list	Firmware name
FLASH_PARTITION_0	bootparam_sa0.srec
FLASH_PARTITION_1	icumx_loader.srec
FLASH_PARTITION_2	cert_header_sa9.srec
FLASH_PARTITION_3	dummy_fw.srec
FLASH_PARTITION_4	cr52_loader.srec
FLASH_PARTITION_5	dummy_rtos.srec
FLASH_PARTITION_6	bl31.srec
FLASH_PARTITION_7	u-boot-elf.srec

Table 5-12. eMMC partition list.

Partition list	Firmware name
eMMC_PARTITION_0	bootparam_sa0.srec
eMMC_PARTITION_1	icumx_loader.srec
eMMC_PARTITION_2	cert_header_sa9.srec
eMMC_PARTITION_3	dummy_fw.srec
eMMC_PARTITION_4	cr52_loader.srec
eMMC_PARTITION_5	dummy_rtos.srec
eMMC_PARTITION_6	bl31.srec
eMMC_PARTITION_7	u-boot-elf.srec

5.9.1. Download bootparam_sa0, cert_header_sa9, icumx_loader, cr52_loader to



QSPI Flash and download dummy_rtos, bl31, u-boot-elf to eMMC.

1. Check FLASH_PARTITION:

```
$vi {Firmware Download Folder}/FlashWriter_V4H.sh  
FLASH_PARTITION="FLASH_PARTITION_0 FLASH_PARTITION_1 FLASH_PARTITION_2  
FLASH_PARTITION_3 FLASH_PARTITION_4"  
eMMC_PARTITION="eMMC_PARTITION_5 eMMC_PARTITION_6  
eMMC_PARTITION_7"
```

2. In Uubuntu terminal.

```
$cd {Firmware Download Folder}  
$sudo ./FlashWriter_V4H.sh -a 1
```

5.9.2. Download bootparam_sa0, cert_header_sa9, icumx_loader, cr52_loader,

dummy_rtos, bl31, u-boot-elf to QSPI Flash.

1. Check FLASH_PARTITION:

```
$vi {Firmware Download Folder}/FlashWriter_V4H.sh  
FLASH_PARTITION="FLASH_PARTITION_0 FLASH_PARTITION_1 FLASH_PARTITION_2  
FLASH_PARTITION_3 FLASH_PARTITION_4 FLASH_PARTITION_5  
FLASH_PARTITION_6 FLASH_PARTITION_7"
```

2. In Uubuntu terminal.

```
$cd {Firmware Download Folder}  
$sudo ./FlashWriter_V4H.sh -af 1
```

5.9.3. Download bootparam_sa0, cert_header_sa9, icumx_loader, cr52_loader,

dummy_rtos, bl31, u-boot-elf to eMMC.

1. Check FLASH_PARTITION:

6. Boot

When Flash Writer has downloaded firmware, ICUMX can select which boot device:

Table 6-1. Select boot device.

SW28				Selection of boot device
Pin5	Pin6	Pin7	Pin8	
ON	ON	OFF	ON	HyperFlash boot at 160MHz using DMA
ON	ON	OFF	OFF	HyperFlash boot at 80MHz using DMA
ON	OFF	ON	ON	Serial Flash boot at single read 40MHz using DMA
ON	OFF	OFF	ON	Serial Flash boot at DDR read 80MHz using DMA
ON	OFF	OFF	OFF	Octal SPI Flash 160/80MHz using DMA
OFF	ON	OFF	ON	HyperFlash at 160MHz (320Mbps) using XIP mode
OFF	ON	OFF	OFF	HyperFlash at 80MHz using XIP mode
OFF	OFF	ON	OFF	eMMC boot at 50MHz x8 bus widths using DMA
OFF	OFF	OFF	OFF	SCIF/HSCIF download mode
Other than above				Setting prohibited

6.1. Step of ICUMX boot from QSPI flash on SDR 40MHz:

6.1.1. Power off.

6.1.2. Change SW28:

Table 6-2. Setting QSPI flash boot at 40MHz in SDR mode using DMA.

	Pin1	Pin2	Pin3	Pin4	Pin5	Pin6	Pin7	Pin8
SW28	OFF	ON	OFF	OFF	ON	OFF	ON	ON

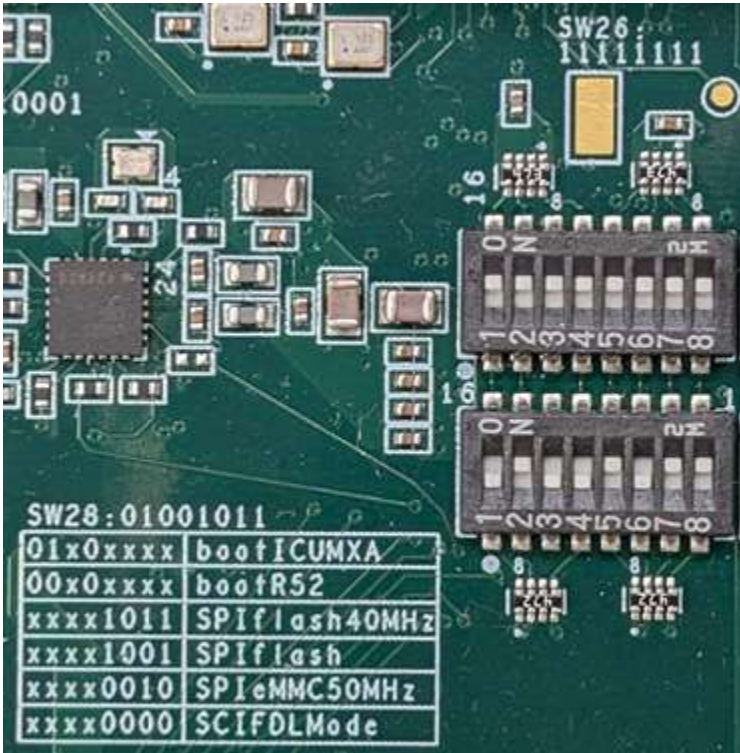


Figure 6-1. QSPI flash boot at 40MHz in SDR mode using DMA of SW28.

6.1.3. Power on.

6.1.4. The firmware is executing and loading:


```

/dev/ttyUSB0 - PuTTY
N:ICUMX Loader Rev.0.22.0
N:Built : 06:55:18, Jun 24 2024
N:PRR is R-Car V4H Ver2.0
N:Boot device is QSPI Flash(40MHz)
N:LCM state is CH
N:Normal boot(ICUMX)
N:===== content cert info =====
destination address:0xfde30000
physical destination address:0xeb230000
source address:0x08240000
size:0x00001000
N:===== content of SA9 =====
address:0xeb231000 size:0x00009000
N:===== Cx IPL image load info =====
load address = 0xE6300000
image size = 0x00060000
source address = 0x08480000
N:===== Secure FW image load info =====
load address = 0xEB240000
image size = 0x000A0000
source address = 0x08280000
N:eMMC Transfer mode: HS400
N:CR52 Loader Program Rev.0.19.0
N:Built : 13:54:01, Jun 24 2024
N:PRR is R-Car V4H Ver.2.0
N:DDR6400(rev.0.40rc5)
N:[COLD_BOOT]
N:..0
N:QoS setting(v8.0.1)
N:DRAM refresh interval 1.91 usec
N:Periodic Write DQ Training
N:===== RTOS image load info =====
N:load address = 0xe2100000
image size = 0x01000000
source address = (p:1)0x001a0000
N:===== CA Program #1 image load info =====
N:load address = 0x46400000
image size = 0x00100000
source address = (p:1)0x011a0000
N:===== CA Program #2 image load info =====
N:load address = 0x50000000
image size = 0x00200000
source address = (p:1)0x012a0000
NOTICE: BL3-1 : Rev.0.10.0
NOTICE: BL31: v2.7(release):
NOTICE: BL31: Built : 03:47:07, Jan 31 2023

N:Dummy FW Program
N:Dummy FW Program boot end

U-Boot 2022.01 (May 27 2023 - 01:48:55 +0000)

CPU: Renesas Electronics R8A779G0 rev 2.0
Model: Renesas White Hawk CPU and Breakout boards based on r8a779g0
DRAM: 7.9 GiB
RAM Configuration:
Bank #0: 0x048000000 - 0x0bfffffff, 1.9 GiB
Bank #1: 0x480000000 - 0x4fffffff, 2 GiB
Bank #2: 0x600000000 - 0x6fffffff, 4 GiB

MMC: mmc@ee140000: 0
Loading Environment from MMC... OK
In: serial@e6540000
Out: serial@e6540000
Err: serial@e6540000
Net:

Dummy RTOS Program
Dummy RTOS Program boot end
eth0: ethernet@eb800000
Hit any key to stop autoboot: 2 █
  
```

ICUMX IPL executing

Boot device is QSPI flash(40MHz)

Load SA9 from QSPI flash

Load CX 2nd IPL from QSPI flash

Load Dummy FW from QSPI flash

CX 2nd IPL executing

Load Dummy RTOS from eMMC

Load bl31 from eMMC

Load U-boot from eMMC

bl31 executing

Dummy FW executing

U-boot executing

Dummy RTOS executing

Figure 6-2. The firmware is loading from QSPI flash at 40MHz and executing.

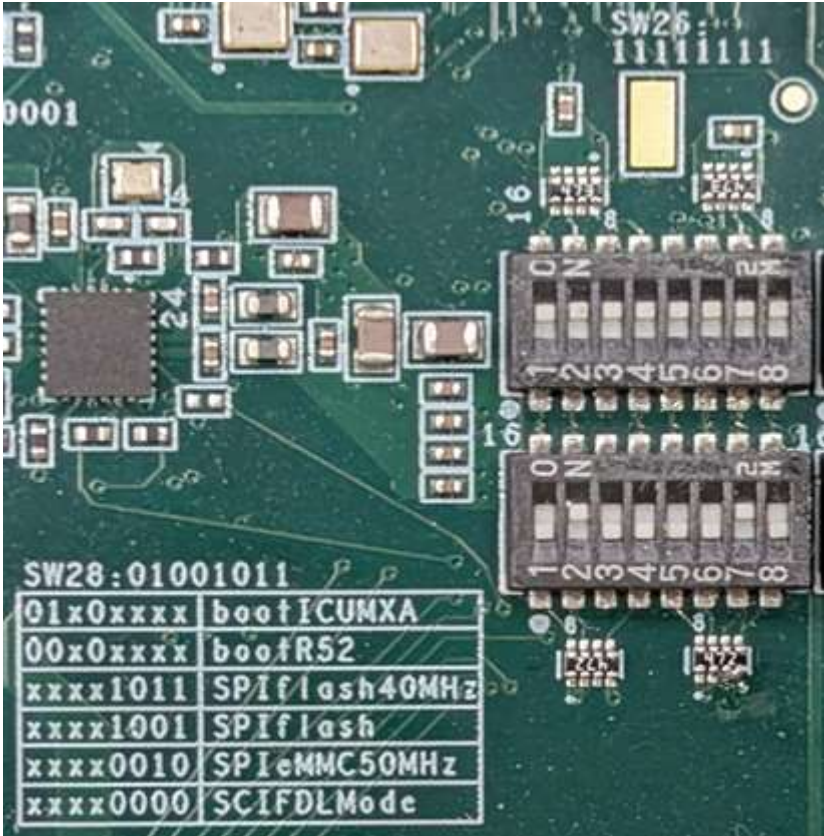


Figure 6-4. eMMC boot 50MHz x8 bus widths using DMA of SW28.

6.2.3. Power on.

6.2.4. The firmware is executing and loading:

```

/dev/ttyUSB0 - PuTTY
N:ICUMX Loader Rev.0.22.0
N:Built : 04:52:48, Jun 27 2024
N:PRR is R-Car V4H Ver2.0
N:Boot device is eMMC(50MHz x8)
N:LCM state is CM
N:Normal boot(ICUMX)
N:===== content cert info =====
destination address:0xfde30000
physical destination address:0xeb230000
source address:(p:1)0x00040000
size:0x00001000
N:===== content cert =====
address:0xeb231000 size:0x00009000
N:===== CX IPL image load info =====
load address = 0xe6300000
image size = 0x00060000
source address = (p:1)0x00100000
N:===== Secure FW image load info =====
load address = 0xeb240000
image size = 0x000a0000
source address = (p:1)0x00080000
N:Dummy FW Program
N:Dummy FW Program boot end
N:eMMC transfer mode: HS400
N:CR52 Loader Program Rev.0.19.0
N:Built : 11:38:48, Jun 27 2024
N:PRR is R-Car V4H Ver.2.0
N:DDR6400(rev.0.40rc5)
N:[COLD_BOOT]
N:.0
N:QoS setting(v8.0.1)
N:DRAM refresh interval 1.91 usec
N:Periodic Write DR Training
N:===== RTOS image load info =====
N:load address = 0xe2100000
image size = 0x01000000
source address = (p:1)0x001a0000
N:===== CA Program #1 image load info =====
N:load address = 0x46400000
image size = 0x00100000
source address = (p:1)0x011a0000
N:===== CA Program #2 image load info =====
N:load address = 0x50000000
image size = 0x00200000
source address = (p:1)0x012a0000
NOTICE: BL3-1 : Rev.0.10.0
NOTICE: BL31: v2.7(release):
NOTICE: BL31: Built : 03:47:07, Jan 31 2023
U-Boot 2022.01 (May 27 2023 - 01:48:55 +0000)
CPU: Renesas Electronics R8A779G0 rev 2.0
Model: Renesas White Hawk CPU and Breakout boards based on r8a779g0
DRAM: 7.9 GiB
RAM Configuration:
Bank #0: 0x048000000 - 0x0bfffffff, 1.9 GiB
Bank #1: 0x480000000 - 0x4fffffff, 2 GiB
Bank #2: 0x600000000 - 0x6fffffff, 4 GiB
MMC: mmc@ee140000: 0
Loading Environment from MMC... OK
In: serial@e6540000
Out: serial@e6540000
Err: serial@e6540000
Net:
Dummy RTOS Program
Dummy RTOS Program boot end
eth0: ethernet@b800000
Hit any key to stop autoboot: 2
  
```

ICUMX IPL executing

Boot device is eMMC

Load SA9 from eMMC

Load CX 2nd IPL from eMMC

Load Dummy FW from eMMC

Dummy FW executing

CX 2nd IPL executing

Load Dummy RTOS from eMMC

Load bl31 from eMMC

Load U-boot from eMMC

bl31 executing

U-boot executing

Dummy RTOS executing

Figure 6-5. The firmware is loading from eMMC at 50MHz and executing.

6.2.5. The firmware loading sequence:

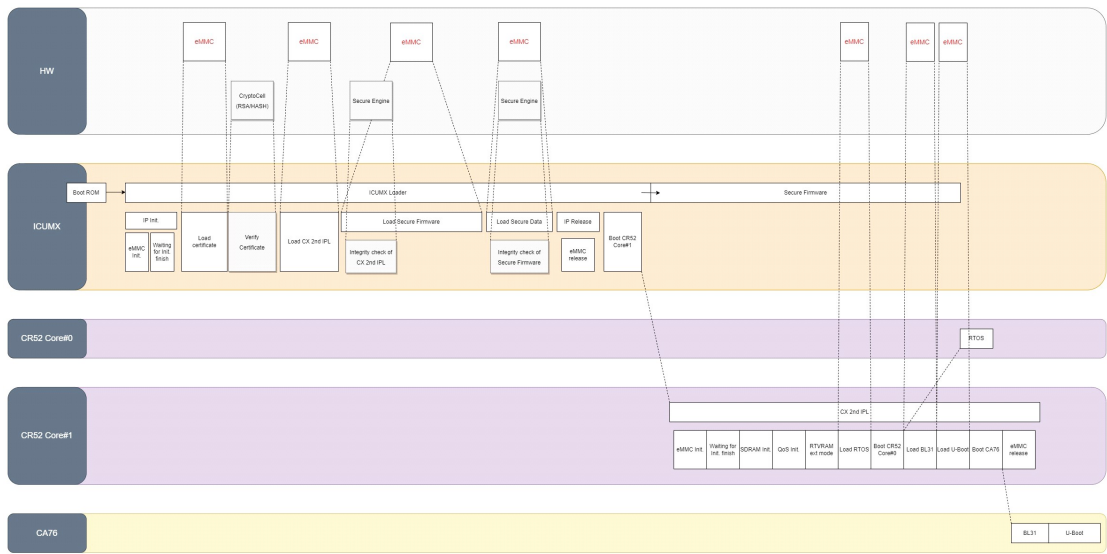


Figure 6-6. The firmware loading sequence.

6.3. Step of ICUMX boot from QSPI flash on DDR 80MHz.

6.3.1. Power off.

6.3.2. Change SW28:

Table 6-4. Setting QSPI flash boot at 80MHz in DDR mode using DMA.

	Pin1	Pin2	Pin3	Pin4	Pin5	Pin6	Pin7	Pin8
SW28	OFF	ON	OFF	OFF	ON	OFF	OFF	ON

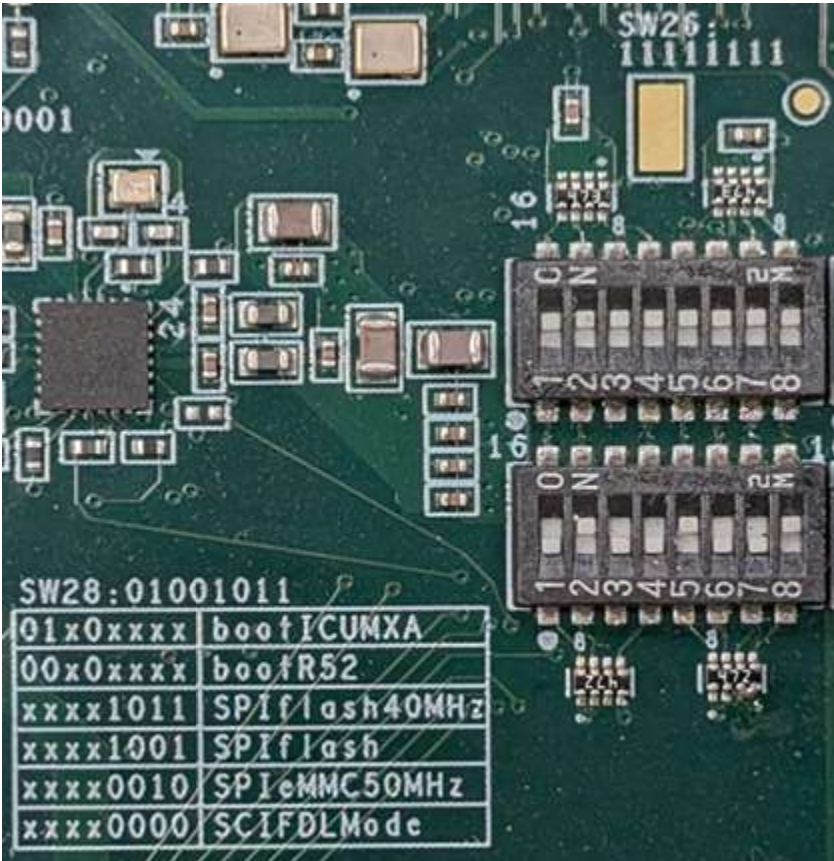


Figure 6-7. QSPI flash boot at 80MHz in DDR mode using DMA of SW28.

6.3.3. Power on.

6.3.4. The firmware is executing and loading:

```

/dev/ttyUSB0 - PuTTY
N:ICUMX Loader Rev.0.22.0
N:Built : 10:39:45, Jun 27 2024
N:PRR is R-Car V4H Ver2.0
N:Boot device is QSPI Flash(80MHz)
N:LCM state is CM
N:Normal boot(ICUMX)
N:===== content cert info =====
destination address:0xfde30000
physical destination address:0xeb230000
source address:0x08240000
size:0x00001000
N:===== content of SA9 =====
address:0xeb231000 size:0x00009000
N:===== CX IPL image load info =====
load address = 0xE6300000
image size = 0x00060000
source address = 0x08480000
N:===== Secure FW image load info =====
load address = 0xEB240000
image size = 0x000A0000
source address = 0x08280000

NN::DCuRn5m2y LFolwa dPferro gPrRaong
oNm: DRuemvm.y0 .F1W9 .P0r
gNr:aBmu ibloto t: eIn7d:
9:58, Jun 27 2024 3
N:PRR is R-Car V4H Ver.2.0
N:DDR6400(rev.0.40rc5)
N:[COLD_BOOT]
N:...0
N:QoS setting(v8.0.1)
N:DRAM refresh interval 1.91 usec
N:Periodic Write DQ Training
N:===== RTOS image load info =====
load address = 0xe2100000
image size = 0x01000000
source address = 0x08500000
N:===== CA Program #1 image load info =====
load address = 0x46400000
image size = 0x00100000
source address = 0x08600000
N:===== CA Program #2 image load info =====
load address = 0x50000000
image size = 0x00200000
source address = 0x08700000
NOTICE: BL3-1 : Rev.0.10.0
NOTICE: BL31: v2.7(release):
NOTICE: BL31: Built : 03:47:07, Jan 31 2023

U-Boot 2022.01 (May 27 2023 - 01:48:55 +0000)

CPU: Renesas Electronics R8A779G0 rev 2.0
Model: Renesas White Hawk CPU and Breakout boards based on r8a779g0
DRAM: 7.9 GiB
RAM Configuration:
Bank #0: 0x048000000 - 0x0bfffffff, 1.9 GiB
Bank #1: 0x480000000 - 0x4fffffff, 2 GiB
Bank #2: 0x600000000 - 0x6fffffff, 4 GiB
MMC: mmc@ee140000: 0
Loading Environment from MMC...
Dummy RTOS Program
Dummy RTOS Program boot end
UK
In: serial@e6540000
Out: serial@e6540000
Err: serial@e6540000
Net: eth0: ethernet@e6800000
Hit any key to stop autoboot: 2
  
```

ICUMX IPL executing

Boot device is QSPI flash(80MHz)

Load SA9 from QSPI flash

Load CX 2nd IPL from QSPI flash

Load Dummy FW from QSPI flash

Dummy FW executing.
CX 2nd IPL executing.
(Two parts of the message are mixed.)

Load Dummy RTOS from QSPI flash

Load bl31 from QSPI flash

Load U-boot from QSPI flash

bl31 executing

U-boot executing

Dummy RTOS executing

Figure 6-8. The firmware is loading from QSPI flash on DDR 80MHz and executing.

6.3.5. The firmware loading sequence:

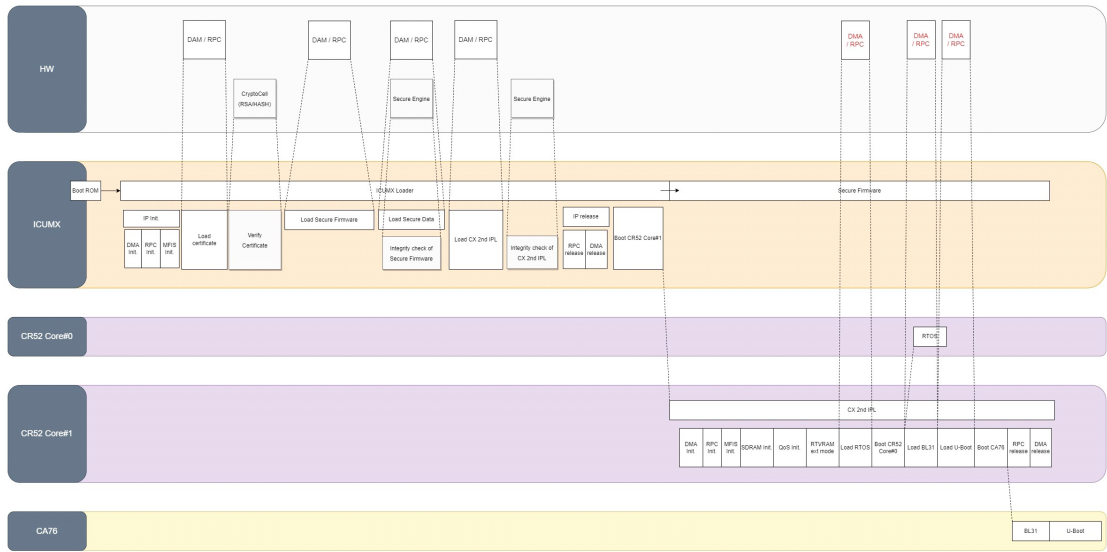


Figure 6-9. The firmware loading sequence.

7. Flash Writer

The Flash writer transfers some of binary images from Host PC via SCIF and writes the binary images to QSPI Flash and eMMC (hereafter referred to as “external flash memory”)

7.1. The Flash writer has the following functions:

- Read the QSPI Flash memory data.
- Write images to the QSPI Flash memory.
- Erase the QSPI Flash memory.
- Display the CID/CSD/EXT_CSD registers of eMMC.
- Modify the EXT_CSD registers of eMMC.
- Read the eMMC memory data.
- Write images to the user data area of eMMC.
- Display the command help.

7.2. Describe the detail of the SCIF download image as follows:

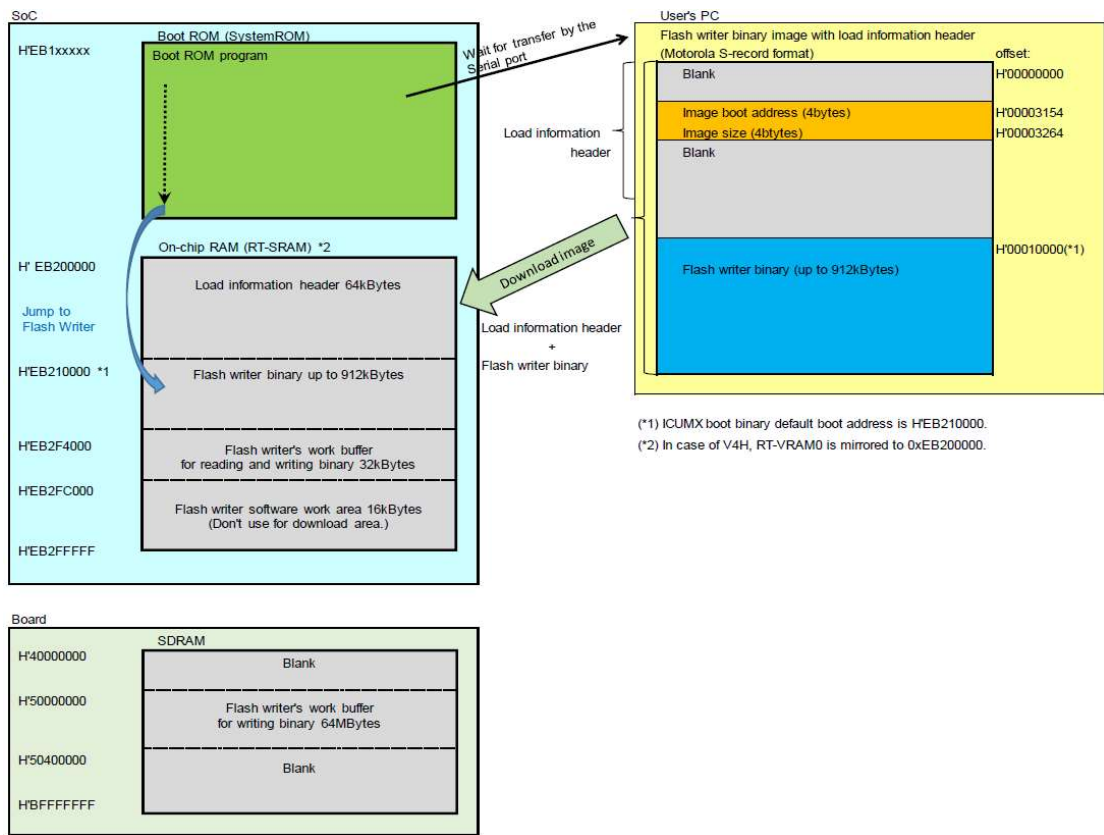


Figure 7-1. Detail of the SCIF download image.

7.3. Command specification

The following table shows the command list:

Table 7-1. Command list.

Command	Description
XRS	Read the QSPI Flash memory data.
XLS2	Write S-record(.srec) format images to the QSPI Flash memory.
XLS3	Write raw binary images to the QSPI Flash memory.
XCS	Erase the QSPI Flash memory.
EM_DCID	Display the CID registers of eMMC.
EM_DCSD	Display the CSD registers of eMMC.
EM_DECSD	Display the EXT_CSD registers of eMMC.
EM_SECS	Modify the EXT_CSD registers of eMMC.
EM_R	Read the eMMC memory data.
EM_W	Write S-record(.srec) format images to the user data area of eMMC and the boot partition of eMMC.

EM_WB	Write raw binary images to the user data area of eMMC and the boot partition of eMMC.
EM_E	Erase the user data area of eMMC, and the boot partition of eMMC.
H	Display the command help.

7.4. FlashWriter_V4H.sh

FlashWriter_V4H.sh can input the command automatically.

Table 7-2. FlashWriter_V4H.sh command list.

Feature	Example
Download all firmware to flash part.1 and eMMC part.1.	\$/FlashWriter_V4H.sh -a 1
Download all firmware to flash part.1.	\$/FlashWriter_V4H.sh -af 1
Read firmware by hex format from flash part.1, begin 0x08400000, length=0x10 bytes.	\$/FlashWriter_V4H.sh -fr 1 08400000 10
Erase flash part.1.	\$/FlashWriter_V4H.sh -fe 1
Download all firmware to eMMC part.1.	\$/FlashWriter_V4H.sh -ae 1
Read firmware by hex format from eMMC part.1, begin sector=0x0080, sector number=0x1.	\$/FlashWriter_V4H.sh -er 1 0080 1
Erase eMMC part.1.	\$/FlashWriter_V4H.sh -ee 1

7.4.1. Download bootparam_sa0, cert_header_sa9, icumx_loader, cr52_loader to

QSPI Flash and download dummy_rtos, bl31, u-boot-elf to eMMC.

Command format:

```
$/FlashWriter_V4H.sh -a [QSPI Flash and eMMC partition1~3]
```

Example:

```
$/FlashWriter_V4H.sh -a 1
```

7.4.2. Download bootparam_sa0, cert_header_sa9, icumx_loader, cr52_loader,

dummy_rtos, bl31, u-boot-elf to QSPI Flash.

Command format:

```
$/FlashWriter_V4H.sh -af [QSPI Flash partition1~3]
```

Example:

```
$/FlashWriter_V4H.sh -af 1
```

7.4.3. Download bootparam_sa0, cert_header_sa9, icumx_loader, cr52_loader,

dummy_rtos, bl31, u-boot-elf to eMMC.

Command format:

```
$/FlashWriter_V4H.sh -ae [eMMC partition0~2]
```

Example:

```
$/FlashWriter_V4H.sh -ae 1
```

7.4.4. Read QSPI flash memory.

Command format:

```
$/FlashWriter_V4H.sh -fr [QSPI Flash partition1~3] [start address] [length]
```

Example:

```
$/FlashWriter_V4H.sh -fr 1 08400000 10
```

7.4.5. Read eMMC memory.

Command format:

```
$/FlashWriter_V4H.sh -er [eMMC partition0~2] [start sector] [sector number]
```

Example:

```
$/FlashWriter_V4H.sh -er 1 0 10
```

7.4.6. Erase QSPI flash memory.

Command format:

```
./FlashWriter_V4H.sh -fe [QSPI Flash partition1~3]
```

Example:

```
./FlashWriter_V4H.sh -fe 1
```

7.4.7. Erase eMMC memory.

Command format:

```
./FlashWriter_V4H.sh -ee [eMMC partition0~2]
```

Example:

```
./FlashWriter_V4H.sh -ee 1
```

8. References

The following table shows the document related to this function.

Table 8-1. Related Document.

Number	Issue	Title	Version
[1]	Renesas Electronics	ICUMX IPL for R-Car Gen4 User’s Manual: Software	Rev.0.21.1
[2]	Renesas Electronics	CX 2nd IPL for R-Car Gen4 User’s Manual: Software	Rev.0.17.2
[3]	Renesas Electronics	R-Car Gen4 Flash writer sample software User’s Manual: Software	Rev.0.13.0
[4]	Renesas Electronics	R-CarV4H_V4M_V3H_V3M_SDK_StartupGuide.pdf	Rev.0.26
[5]	Green Hills Software	MULTI: Installation	-